

COMPUTER NETWORKS (SECN-328)

I. Course Details

Credit Hours	4 (3+1)
Pre-requisites	-
Course Leader	Mr. Hasan Humayun
Recommended Textbook(s)	1. Data Communications and Networking, Behrouz A. Forouzan, 5th Edition, McGraw-Hill, 2012. 2. Computer Networking: A Top Down Approach, James F. Kurose, Keith Ross, Pearson, 2017
Recommended Reference (Books/Websites/Articles)	Data and Computer Communications, William Stallings, 10 th Edition, Pearson, 2014.

II. Course Learning Outcomes (CLO)

CLOs	Description	Domain	Taxonomy Level	PLOs	Assessment Artifact
CLO-1	Explain the basic network components and data representation in networks to state its main ideas.	Cognitive	2	2	A1, Q1, Mid Term
CLO-2	Make use of Data Link layer services, devices, techniques, and protocols within a network to solve related problems.	Cognitive	3	3	A2, Q2, Midterm, Final Term
CLO-3	Identify features, services, and protocols of Network and Transmission layers by applying acquired knowledge.	Cognitive	3	3	A3, Q3, Project/ Presentation, Final Term
CLO-4	Implement the technologies to create an enterprise-level network with active services.	Cognitive	3	4	CCP

III. Course Assessment

Evaluation Methods	Weight (%)
Quizzes	10
Assignments	10
Presentation/Project	5



Midterm	25
Final Term	50
Total	100

IV. Grading Policy

For students admitted in Fall 2021 and onwards

Grade	A+	A	B+	B	C+	C	D+	D	F
%age	>=90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50
GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00

For students admitted before Fall 2021

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	>=90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	<50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

V. Course Contents

Introduction and protocols architecture, basic concepts of networking, network topologies, layered architecture, physical layer functionality, data link layer functionality, multiple access techniques, circuit switching, and packet switching, LAN technologies, MAC address, networking devices, network layer protocols, IPv4 and IPv6, DHCP, NAT, IP addressing, IP datagram, ICMP, routing protocols, UDP and (TCP), ports and sockets, connection establishment, flow and congestion control, application layer protocols, latest trends in computer networks.

VI. Weekly Breakdown

Week No.	CLO	Topics	Reference
1	CLO 1	Intro to Networks components, Basic concepts of networking. Network topologies, network layers, internet standards, and architecture.	Chapter 1
2		The concept of layered architecture modeling. TCP/IP protocol suite, comparison, protocols and OSI.	Chapter 2
3		The distinction between Data and Signals of Analog and Digital Transmission Impairments. Network performance measures including throughput, and delays. Data vs. signaling rates. Channel bandwidth and capacity, delay product.	Chapter 3
4		Switching Technologies: Circuit Switching, Packet Switching, datagram, and Virtual Circuit Switching: Phases, efficiency, and delays.	Chapter 8

5	CLO 2	Data link layer (DLL), Nodes and links, services, framing, flow, error, congestion control, data link addressing, ARP, and communication using DLL.	Chapter 9
6		Error Detection and Correction, Types of Error, Redundancy, Error Detection, Parity, checksum, VRC, LRC, cyclic codes, Detection versus Correction, Forward Error Correction (Hamming codes, Hamming distance) versus Retransmission	Chapter 10
7		Data link control services, Flow, and error control (simple protocol & stop N wait protocol), PAP and CHAP Protocols, Go back N (GBN) and Selective Reject.	Chapter 11 & Chapter 23
8		Multiple Control Access: Random access protocols (Aloha, CSMA, CSMA/CD, CA), Wireless hidden and exposed node problems, and Controlled access protocols (Reservation, Polling, and Token Passing)	Chapter 12
9		Midterm Exam	
10		Ethernet, Evolution, Standard Ethernet addressing, access method Fast Ethernet Access, Mac sub-layer.	Chapter 13
11		WLANs: architecture, characteristics, 802.11 sublayers, Exposed and hidden node problem.	Chapter 15
12	CLO3	Connecting devices: Hub, Switch and Router, Virtual Lans, Spanning tree Algorithm. IP addressing, classful, classless routing, local and global addresses including APIPA.	Chapter 17 & Chapter 18
13		IP addressing format, fragmentation, ICMP v4: error reporting and query messages.	Chapter 19
14		Routing: Least cost, Routing Algorithms: Distance vector and link state, RIP and RIP messages, algorithm, and timers.	Chapter 20
15		UDP Error control and Flow control, congestion. TCP Features, segments, connections, pushing, handshaking, windows in TCP.	Chapter 24
16		Flow control: opening and closing, silly window syndrome. Error Control: acknowledgments, retransmissions. TCP Congestion control, detection, and policies.	Chapter 24